

5.3 The Definite integral

In this section we consider the limit of general Riemann sums as the norm of the partitions of a closed interval $[a, b]$ approaches zero. This limiting process leads us to the definition of the *definite integral* of a function over a closed interval $[a, b]$.

① Definition of the Definite integral

DEFINITION Let $f(x)$ be a function defined on a closed interval $[a, b]$. We say that a number J is the **definite integral of f over $[a, b]$** if

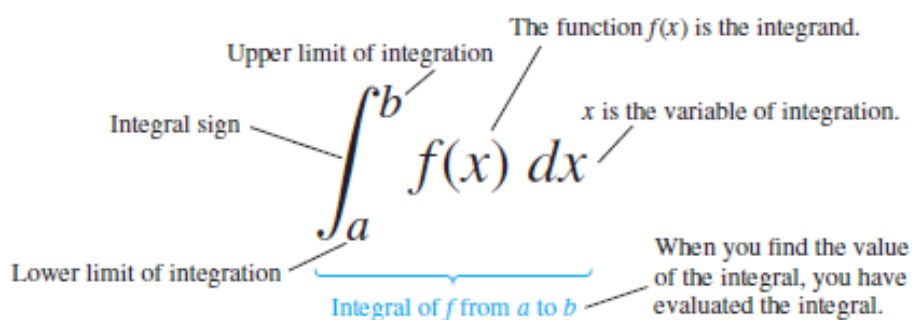
$$J = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k \text{ --- Riemann sum}$$

provided that this limit exists for every partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ and any choice of c_k in $[x_{k-1}, x_k]$.

If the definite integral exists, then instead of writing J we write

$$\int_a^b f(x) dx.$$

Note 1 The integral symbol also has name:



When the definite integral exists, we say that the Riemann sums of f on $[a, b]$ **converge** to the definite integral

$$J = \int_a^b f(x) dx$$

and that f is **integrable** over $[a, b]$.

In the case where the subintervals all have equal width $\Delta x = \frac{b-a}{n}$, the Riemann sums have the form

$$S_n = \sum_{k=1}^n f(c_k) \Delta x_k = \sum_{k=1}^n f(c_k) \frac{b-a}{n},$$

where c_k is chosen in the k th subinterval. If the definite integral exists, then these Riemann sums converge to the definite integral of f over $[a, b]$, so

$$J = \int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \left(\frac{b-a}{n} \right).$$

Note 2 The value of the definite integral of a function over any particular interval depends on the function, not on the letter we choose to represent its independent variable. If we decide to use t or u instead of x , we simply write the integral as

$$\int_a^b f(t) dt \text{ or } \int_a^b f(u) du \text{ instead of } \int_a^b f(x) dx.$$

② Integrable and Nonintegrable Functions

THEOREM 1— integrability of Continuous Functions

If a function f is continuous over the interval $[a, b]$, or if f has at most finitely many jump discontinuities there, then the definite integral

$$\int_a^b f(x) dx$$

exists and f is integrable over $[a, b]$.

Example 1 The function

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational;} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

has no Riemann integral over $[0, 1]$.

Note that between any two numbers there is both a rational number and an irrational

number. If we choose a partition P of $[0, 1]$, and in each subinterval $[x_{k-1}, x_k]$ choose a rational point, say c_k , then $f(c_k) = 1$. Since 1 is the maximum value that f can take anywhere, the upper sum approximation for this choice of c_k 's is

$$U = \lim_{\Delta x_k \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k = \lim_{\Delta x_k \rightarrow 0} \sum_{k=1}^n 1 \cdot \Delta x_k = \lim_{\Delta x_k \rightarrow 0} 1 = 1.$$

As the norm of the partition approaches 0, these upper sum approximations converge to 1 (because each approximation is equal to 1).

On the other hand, we could pick an irrational point c_k , and for this choice $f(c_k) = 0$. Since 0 is the minimum value that f can take anywhere, this choice of c_k gives us the minimum value of f on the subinterval. The corresponding lower sum approximation is

$$L = \lim_{\Delta x_k \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k = \lim_{\Delta x_k \rightarrow 0} \sum_{k=1}^n 0 \cdot \Delta x_k = 0.$$

These lower sum approximations converge to 0 as the norm of the partition converges to 0 (because they each equal 0).

Thus making different choices for the points c_k results in different limits for the corresponding Riemann sums. We conclude that the definite integral of f over the interval $[0, 1]$ does not exist, and that f is not integrable over $[0, 1]$.

③ Properties of Definite Integral

TABLE 5.6 Rules satisfied by definite integrals

1. <i>Order of Integration:</i>	$\int_b^a f(x) dx = -\int_a^b f(x) dx$	A definition
2. <i>Zero Width Interval:</i>	$\int_a^a f(x) dx = 0$	A definition when $f(a)$ exists
3. <i>Constant Multiple:</i>	$\int_a^b kf(x) dx = k \int_a^b f(x) dx$	Any constant k
4. <i>Sum and Difference:</i>	$\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$	
5. <i>Additivity:</i>	$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$	

6. Max-Min Inequality: If f has maximum value $\max f$ and minimum value $\min f$ on $[a, b]$, then

$$(\min f) \cdot (b - a) \leq \int_a^b f(x) dx \leq (\max f) \cdot (b - a).$$

7. Domination: If $f(x) \geq g(x)$ on $[a, b]$ then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$.

If $f(x) \geq 0$ on $[a, b]$ then $\int_a^b f(x) dx \geq 0$. Special case

Example 2 Show that the value of $\int_0^1 \sqrt{1 + \cos x} dx$ is less than or equal to $\sqrt{2}$.

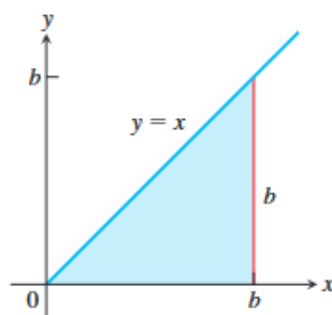
④ Area under the Graph of a Nonnegative Function

DEFINITION If $y = f(x)$ is nonnegative and integrable over a closed interval $[a, b]$, then the **area under the curve $y = f(x)$ over $[a, b]$** is the integral of f from a to b ,

$$A = \int_a^b f(x) dx.$$

Example 3 The area A under $y = x$ over the interval $[0, b]$, $b > 0$ is

$$A = \int_0^b x dx.$$



⑤ Average Value of a Continuous Function

The average value of a nonnegative continuous function f over an interval $[a, b]$ is

$$\text{Average} = \frac{1}{b-a} \int_a^b f(x) dx.$$

Example 4 Find the average value of $f(x) = \sqrt{4 - x^2}$ on $[-2, 2]$.